

A candidate counterexample to Covering powers of cycles with equivalence subgraphs

github.com/graph-theory-AI — machine-generated candidate writeup

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Abstract

Noncomplete powers of cycles have equivalence covering number $\Theta(\log(k+1))$, contradicting the proposed $\Omega(k)$ lower bound. This candidate disproof was generated by GPT-6 Astra in a single model pass for the Graph Theory AI project. It was selected from the campaign because the model marked it `would_publish: true`; it has not been checked by the adversarial referee, a human mathematician, or a novelty search.

1 The problem and the claimed result

This document concerns [Covering powers of cycles with equivalence subgraphs](http://www.open-problemgarden.org/op/covering_powers_of_cycles_with_equivalence_subgraphs) from http://www.open-problemgarden.org/op/covering_powers_of_cycles_with_equivalence_subgraphs.

1.1 Problem statement

Conjecture Given k and n , the graph C_n^k has equivalence covering number $\Omega(k)$.

1.2 Catalog context

Given a graph G , a subgraph H of G is an equivalence subgraph of G if H is a disjoint union of cliques. The equivalence covering number of G , denoted $eq(G)$, is the least number of equivalence subgraphs needed to cover the edges of G . This problem has been studied by various people since the 80s [A]. For line graphs, the equivalence covering number is known to within a constant factor [EGK]. It is therefore tempting to examine the situation for quasi-line graphs and claw-free graphs. Powers of cycles are perhaps the simplest interesting class of claw-free graphs that are not necessarily line graphs. However, even for n very large compared to k , no upper bound is known beyond trivial linear bounds of order $\Theta(k)$. Furthermore, it is not even certain that a nontrivial lower bound (i.e. going to infinity as k goes to infinity) is known. It is possible that this can be related somehow to a known result, but for now it seems at least superficially that this problem is wide open.

1.3 Claimed result

Model verdict: disproved (confidence: high).

Noncomplete powers of cycles have equivalence covering number $\Theta(\log(k+1))$, contradicting the proposed $\Omega(k)$ lower bound.

The model's own caveat: Bibliographic novelty is unverified; an exact formula is obtained when $k+1$ divides n , but not for every n .

2 Model writeup

The text below is the model output verbatim, apart from moving its machine-readable verdict block into the summary above and shifting Markdown heading levels for this document. Mechanical TeX defects and equation tags were normalized where needed for compilation.

2.1 Result

The linear conjecture is false under the definition in the question. The following bounds are uniform in n , so the counterexample does not depend on taking n small relative to k .

Theorem. Let $k \geq 1$, put

$$s = k + 1, \quad r = \lceil \log_2 s \rceil.$$

For every $n \geq 2k + 2$,

$$r + 1 \leq \text{eq}(C_n^k) \leq 2r + 1.$$

Moreover, if $s \mid n$, then

$$\text{eq}(C_n^k) = r + 1.$$

The remaining regime $n \leq 2k + 1$, with $n \geq 3$, consists of complete graphs and has equivalence covering number 1.

In particular, for

$$k = 2^q - 1, \quad n = m2^q, \quad m \geq 2,$$

the exact value is

$$\text{eq}(C_n^k) = q + 1.$$

Here m can be arbitrarily large. For example, taking $m = 2^q$ gives $n/k \rightarrow \infty$, while

$$\frac{\text{eq}(C_n^k)}{k} = \frac{q + 1}{2^q - 1} \rightarrow 0.$$

All equivalence subgraphs below may be made spanning by adding isolated vertices. Edges of the ambient graph between different chosen cliques are simply omitted from that equivalence subgraph.

2.2 1. Binary splitting of a boundary

Write each integer $x \in \{0, \dots, s - 1\}$ using r binary digits, most significant first:

$$x = x_1 x_2 \cdots x_r.$$

For a position $h \in \{1, \dots, r\}$, a binary prefix p of length $h - 1$, and $\varepsilon \in \{0, 1\}$, define

$$I_h^\varepsilon(p) = \{x \in \{0, \dots, s - 1\} : (x_1, \dots, x_{h-1}) = p, x_h = \varepsilon\}.$$

Two elementary properties will be used:

1. If $x \in I_h^1(p)$ and $y \in I_h^0(p)$, then $x > y$.
2. Whenever $x > y$, their first differing binary digit gives an h, p such that

$$x \in I_h^1(p), \quad y \in I_h^0(p).$$

Thus the relation $x > y$ is covered by these binary comparisons, using r levels.

2.3 2. An $(r + 1)$ -cover when $s \mid n$

Suppose $n = ms$, where $m \geq 2$. Partition the cycle into consecutive blocks

$$B_i = \{v_{i,0}, \dots, v_{i,s-1}\}, \quad i \in \mathbb{Z}/m\mathbb{Z}.$$

The cyclic order runs through the local indices $0, \dots, s - 1$ in each block.

Let H_0 be the disjoint union of the cliques on the blocks B_i . These are valid cliques because each block consists of $s = k + 1$ consecutive vertices.

For each binary position h , and for every block i and prefix p , form

$$Q_{i,h,p} = \{v_{i,x} : x \in I_h^1(p)\} \cup \{v_{i+1,y} : y \in I_h^0(p)\}.$$

Let H_h be the union of the complete graphs on these sets, omitting empty sets.

2.3.1 Each $Q_{i,h,p}$ is a clique

Pairs in the same block are adjacent. For a cross-block pair, its local indices satisfy $x > y$. The clockwise distance from $v_{i,x}$ to $v_{i+1,y}$ is

$$s + y - x \leq s - 1 = k.$$

Hence every cross-block pair is also adjacent.

2.3.2 The cliques in H_h are vertex-disjoint

Fix h . A vertex $v_{i,x}$ has one binary prefix p of length $h - 1$.

- If $x_h = 1$, it belongs to $Q_{i,h,p}$.
- If $x_h = 0$, it belongs to $Q_{i-1,h,p}$.

It belongs to no other such set. Consequently, H_h is an equivalence subgraph.

This is the key compatibility: at a given binary level, a vertex participates across its right boundary or its left boundary, but never both.

2.3.3 All edges are covered

An edge of C_n^{s-1} has a cyclic arc of length at most $s - 1$ joining its endpoints. Such an arc cannot cross two block boundaries.

If its endpoints lie in one block, H_0 covers it. Otherwise, orient its short arc from B_i to B_{i+1} , and write the endpoints as $v_{i,x}, v_{i+1,y}$. Adjacency gives

$$s + y - x \leq s - 1,$$

so $x > y$. At their first differing binary digit, the two endpoints belong to the same $Q_{i,h,p}$.

Therefore

$$\text{eq}(C_{ms}^{s-1}) \leq r + 1.$$

Some of these cliques have gaps in the cyclic order. That is allowed: they are cliques, though not necessarily consecutive or maximal cliques.

2.4 3. A uniform upper bound for arbitrary n

We first record the linear version of the preceding construction.

2.4.1 Path-power cover

For every N , the graph on $0, \dots, N - 1$ with edges

$$uv \in E \iff 0 < |u - v| \leq s - 1$$

has an equivalence cover of size at most $r + 1$.

Indeed, partition the linear order into blocks of size s , with the last block possibly shorter. Use the block-clique layer H_0 , and use the same binary construction across every consecutive pair of blocks. At each binary level, the cliques remain vertex-disjoint by the same left/right membership argument.

Every edge lies within one block or crosses one boundary. Across a boundary, the local indices again satisfy $x > y$, so the first differing binary digit covers the edge. Missing vertices in the final block cause no difficulty.

2.4.2 Closing the cycle

Now label the vertices of C_n^{s-1} by $0, \dots, n - 1$, where $n \geq 2s$.

The non-wrapping edges, namely those with ordinary index difference at most $s - 1$, are covered by the preceding path-power construction using $r + 1$ layers.

Every remaining edge wraps across the cut between $n - 1$ and 0 . Consider the disjoint sets

$$A = \{n - s, \dots, n - 1\}, \quad B = \{0, \dots, s - 1\}.$$

Both are cliques. Write their vertices as

$$a_x = n - s + x, \quad b_y = y, \quad 0 \leq x, y < s.$$

A wrap-around edge satisfies

$$n - a_x + b_y = s + y - x \leq s - 1,$$

equivalently $x > y$. Conversely, every pair with $x > y$ is an edge across this boundary.

For each h, p , take the clique

$$\{a_x : x \in I_h^1(p)\} \cup \{b_y : y \in I_h^0(p)\}.$$

For fixed h , these cliques are vertex-disjoint, and the r binary levels cover all wrap-around edges.

Combining the two covers gives

$$\text{eq}(C_n^{s-1}) \leq (r + 1) + r = 2r + 1.$$

2.5 4. A logarithmic lower bound by matrix rank

The following elementary rank argument proves the lower bound, including the exact value in the divisible case.

Let

$$t = \text{eq}(C_n^{s-1}), \quad n \geq 2s.$$

Since $s \geq 2$, this graph has edges and $t \geq 1$.

For each layer $\ell \in \{1, \dots, t\}$, assign distinct real labels to its clique components, and let $c_\ell(v)$ be the label of the component containing v . Then, for distinct vertices,

$$uv \in E(C_n^{s-1}) \iff c_\ell(u) = c_\ell(v) \text{ for some } \ell.$$

Define an $n \times n$ real matrix

$$M_{uv} = \prod_{\ell=1}^t (c_\ell(u) - c_\ell(v)).$$

Thus $M_{uv} = 0$ exactly when $u = v$ or uv is an edge.

2.5.1 Rank upper bound

For $S \subseteq [t]$, let

$$a_S(v) = \prod_{\ell \in S} c_\ell(v),$$

using the empty-product convention. Expanding the product gives

$$M = \sum_{S \subseteq [t]} (-1)^{t-|S|} a_S a_{[t] \setminus S}^T.$$

Each summand has rank at most one, so

$$\text{rank } M \leq 2^t.$$

2.5.2 A nonsingular minor of order $2s$

Use cyclic vertex labels modulo n , and consider the ordered submatrix

$$N_{ij} = M_{i, (s+j) \bmod n}, \quad 0 \leq i, j < 2s.$$

Its row indices are distinct and its column indices are distinct, since $2s \leq n$.

If $i > j$, then

$$1 - s \leq s + j - i \leq s - 1.$$

Therefore the corresponding vertices are equal or have cyclic distance at most $s - 1$, and

$$N_{ij} = 0.$$

On the diagonal, the two vertices have cyclic distance exactly s , because $n \geq 2s$. They are distinct and nonadjacent, so

$$N_{ii} \neq 0.$$

Hence N is upper triangular with nonzero diagonal. It follows that

$$2s = \text{rank } N \leq \text{rank } M \leq 2^t.$$

Consequently,

$$t \geq \lceil \log_2(2s) \rceil = 1 + \lceil \log_2 s \rceil = r + 1.$$

Together with the construction in Section 2, this proves

$$\text{eq}(C_{ms}^{s-1}) = 1 + \lceil \log_2 s \rceil \quad (m \geq 2).$$

2.6 Conclusion and scope

The bounds establish

$$\text{eq}(C_n^k) = \Theta(\log(k + 1))$$

uniformly over all positive k and all $n \geq 2k + 2$. Thus the proposed $\Omega(k)$ lower bound fails even when n is arbitrarily large relative to k .

No unproved conjecture or external theorem is used. The exact value for every nondivisible n is not determined here. I have also not verified whether this binary construction and rank argument already appear in the literature; the conclusion is a self-contained disproof of the stated conjecture, not a claim of historical novelty.

A Provenance and review status

UNREFEREED MODEL OUTPUT. This document was selected solely because GPT-6 Astra labelled its own result `would_publish: true` and returned `proved` or `disproved`. It has not passed the adversarial LLM referee used for the earlier Sol campaign, has not been checked by a human mathematician, and has not been checked for novelty. Treat every mathematical and bibliographic claim as unverified.

Catalog id	covering_powers_of_cycles_with_equivalence_subgraphs
Catalog entry	Covering powers of cycles with equivalence subgraphs
Source corpus	opg
Campaign leg	OpenProblemGarden first pass (attacks_opg)
Source paper / entry	http://www.openproblemgarden.org/op/covering_powers_of_cycles_with_equivalence_subgraphs
Model verdict	disproved (confidence: high)
Selection criterion	<code>would_publish: true</code> and verdict <code>proved</code> or <code>disproved</code>
Model	<code>gpt-6-astra</code> , reasoning effort <code>max</code> , mode= <code>pro</code> , flex service tier
Original artifact	attacks_opg/covering_powers_of_cycles_with_equivalence_subgraphs/
Independent review	None
Model’s caveats	Bibliographic novelty is unverified; an exact formula is obtained when $k+1$ divides n , but not for every n .

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